

Regional Profitability Analysis: Identifying Drivers of Low Profit in the Central Region

Project Overview

This project analyzes the Sample Superstore retail dataset to investigate why the Central region generates lower profitability than the other regions. Using Power Query for data cleaning, SQL Server for analysis, and Tableau for visualization, the project identifies the key drivers of poor financial performance and provides business recommendations.

Business Problem

The company experienced lower profitability in the Central region compared with other regions. This analysis investigates the factors contributing to this performance gap and identifies the categories and subcategories responsible for profit losses.

Business Questions

- Why does the Central region generate lower profit compared with other regions?
- Which product categories contribute most to profit losses?
- Which subcategories within Furniture are causing negative profitability?
- Are high sales categories also generating strong profit margins?

Objective of the Analysis

The objective of this project is to analyze regional, category, and subcategory performance to identify the main drivers of low profitability and provide data-driven recommendations for improving business performance.

Dataset

The dataset used in this project is the Sample Superstore dataset, a publicly available retail sales dataset commonly used for data analytics practice and visualization projects.

Dataset Limitations

The dataset provides sales and profit information but does not include detailed cost components such as production costs, shipping costs, operational expenses, or other business-related expenses. Therefore, the analysis can identify which products and categories are associated with lower profitability, but it cannot determine the exact cost factors causing profit losses.

Tools Used

- Power query
- SQL-Server
- Tableau

Data Preparation

- Removed duplicate records
- Checked missing values
- Verified data types
- Trimmed text fields
- Removed unnecessary columns
- Validated data consistency
- Imported cleaned dataset into SQL Server

SQL Analyses

SQL was used to calculate:

- Regional sales and profit
- Category and subcategory performance
- Regional comparisons
- Average Discounts
- Profit margins

Analysis & Findings

Regional Performance Comparison

The West region recorded the highest sales and profit, making it the strongest-performing region overall. The East region ranked second in both sales and profit.

Although the Central region ranked third in sales, it generated the lowest profit among all regions. In contrast, the South region recorded the lowest sales but ranked third in profit, outperforming the Central region.

The Central region generated approximately **\$109,517.98** more in sales than the South region. However, the South region earned approximately **\$7,043.07** more in profit than the Central region. This demonstrates that higher sales do not necessarily translate into higher profitability.

East Region – Category Performance Analysis

Technology is the strongest-performing category in the East region. It generated **\$264,973.98** in sales and **\$47,462.04** in profit, making it the top category in both revenue and profitability.

Furniture ranked second in sales with **\$208,009.83**, but generated only **\$3,058.22** in profit, placing it third in profitability. This suggests that although furniture sales are relatively high, the category contributes very little profit.

Office Supplies generated **\$205,516.06** in sales, slightly below Furniture, but earned **\$41,014.58** in profit, making it the second most profitable category.

West Region – Category Performance Analysis

Furniture generated the highest sales in the West region but produced only a small profit, indicating a low profit margin. In contrast, Office Supplies generated lower sales than Furniture but achieved the highest profit among all categories.

This suggests that office supplies is considerably more profitable, while furniture profitability appears to be lower in sub-categories with higher discount levels.

Central Region – Category Performance Analysis

Technology generated the highest sales and profit in the Central region, making it the strongest-performing category. Although the sales figures for Technology, Office Supplies, and Furniture are relatively close, their profitability differs significantly. Furniture generated sales comparable to the other categories but recorded a loss of **\$2,871.05**, while Technology and Office Supplies remained profitable.

The Central region also recorded the highest average discount (**0.24**), compared with the West region, which had the lowest average discount (**0.11**). This suggests that heavier discounting is a major factor contributing to the region's lower profitability.

A more detailed analysis shows that **Tables, Furnishings, and Bookcases** subcategories were the primary contributors to Furniture's negative performance. These products were frequently sold with discounts ranging from **23% to 40%**, substantially reducing profitability.

South Region – Category Performance Analysis

Technology was the strongest performing category in the South region, generating the highest sales and profit. Although Office Supplies had approximately \$23,000 lower sales than Technology, it generated nearly the same profit, with a difference of only around \$5. This indicates that Office Supplies was more profitable relative to its sales and more efficient at converting revenue into profit. Furniture showed the weakest performance, ranking third in both sales and profit.

Profit Margin Analysis

The profit margin analysis shows that the West region achieved the highest profit margin (**14.94%**), while the Central region recorded the lowest (**7.92%**). This indicates that the Central region not only generated lower total profit but was also less efficient at converting sales into profit. Higher discount levels and a less profitable product mix appear to be key contributing factors.

Tableau Dashboard

The Tableau dashboard was created to visualize regional performance, category profitability, average discount levels, and profit margin comparisons. The dashboard includes visualizations comparing sales and profit across regions, category performance, and the relationship between discounts and profitability.

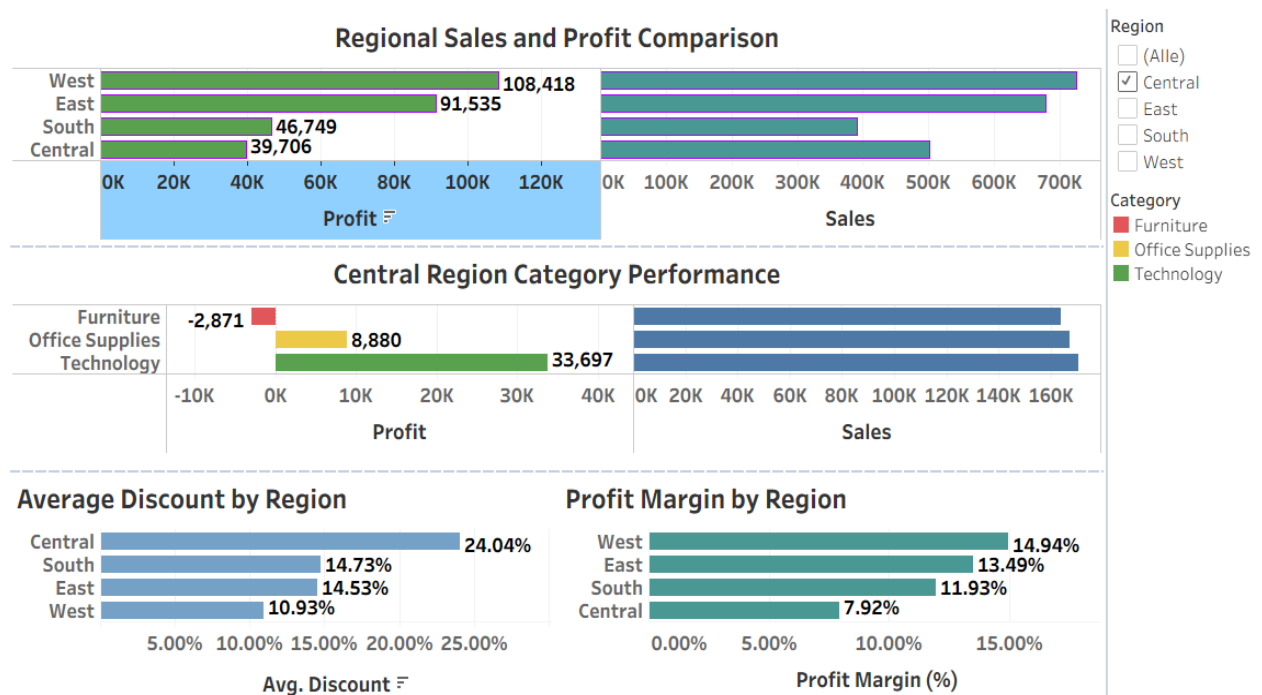


Figure 1. Tableau Dashboard : Regional Sales, Profit, Discount and Profit Margin Analysis

Key Findings

- The West region achieved the highest sales and profit.
- The Central region had the lowest profit margin despite relatively high sales.
- High discount levels in the Central region were associated with lower profitability.
- Furniture subcategories such as Tables, Bookcases, and Furnishings contributed most to the losses

Business Recommendations

- Review discount policies in the Central region.
- Reduce discounts on Tables, Bookcases, and Furnishings.
- Promote Technology products because they generate consistently high profits.
- Monitor Furniture profit margins monthly.
- Investigate supplier or logistics costs for low-performing Furniture products.

Summary

This project analyzed Superstore sales data to identify why the Central region generated lower profitability than the other regions. Using Power Query for data cleaning, SQL for analysis, and Tableau for visualization, the project found that high discounts on Furniture products were the primary cause of low profit. Based on these findings, business recommendations were provided to support more profitable pricing and discount strategies.